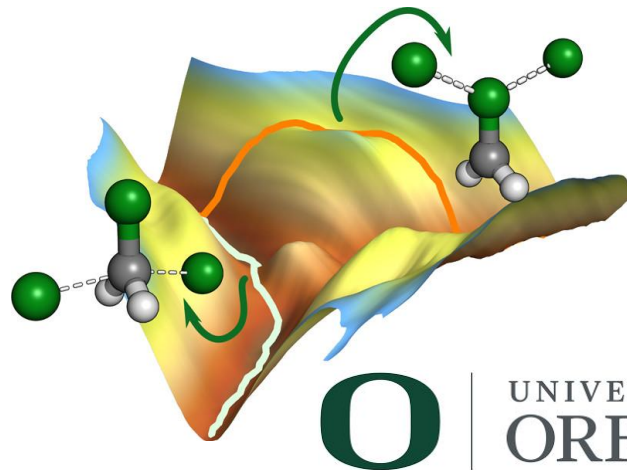
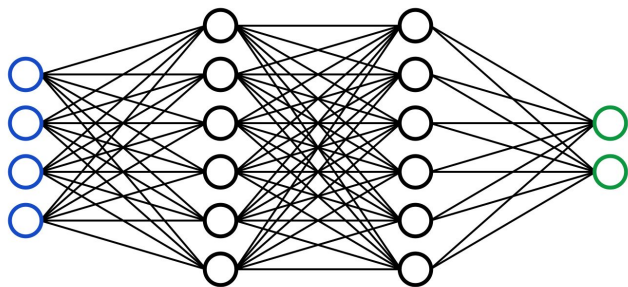


Machine Learning for Chemistry

Winter 2025

Topic 1: Introduction to Machine Learning

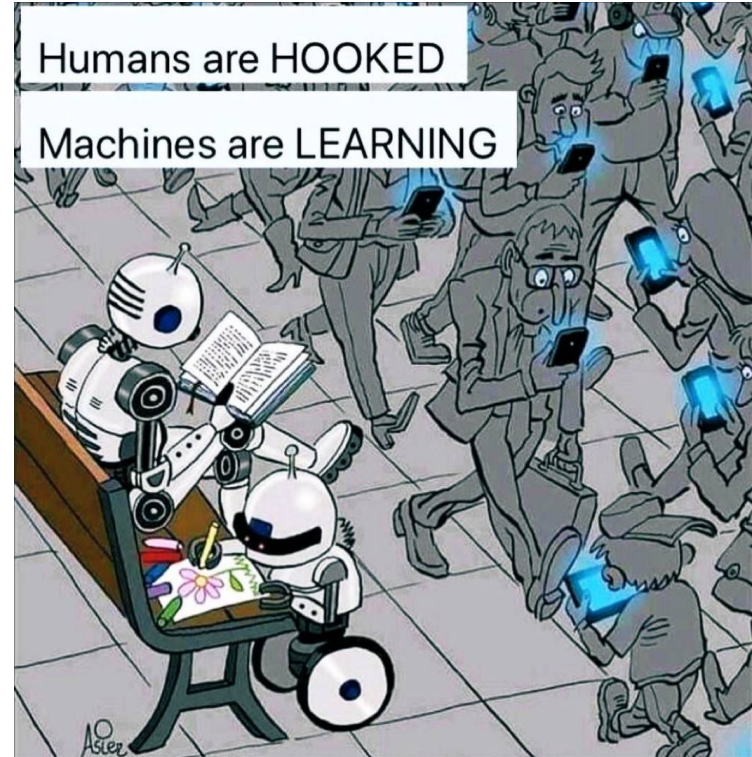


Instructor: Dr. Dhiman Ray

Assistant Professor, Department of Chemistry and Biochemistry

Overview of Topics

- Course Logistics
- What is Machine Learning?
- Why use Machine Learning in Chemistry?
- Different types of Machine Learning
- Basic introduction to Python



Course Logistics: Key Considerations

Goal: To provide a fundamental understanding of the key concepts of Machine Learning models to help you apply them effectively in your research in Chemistry.

This course is not meant to teach you to code/use chatbots!

Basic background of undergraduate level mathematics and python programming will be necessary. Tutorials and guidance will be provided along the way.

Each two hour class will have a lecture and a hands-on in-class tutorial. We will use Google Colab for the tutorials. Please bring your own laptop.

Course Logistics: Topics

Topic 1 Introduction (Today)	Topic 7 Clustering
Topic 2 Challenges in Machine Learning	Topic 8 Deep Learning
Topic 3 Regression	Advanced Topic 1 Convolutional Neural Networks
Topic 4 Classification	Advanced Topic 2 Graph Neural Networks
Topic 5 Tree-Based Methods	Advanced Topic 3 Generative Machine Learning
Topic 6 Dimensionality Reduction	Project and Research Paper Reading

Course Logistics: Evaluation

Homework: (40%) (Weekly) Theory + Programming

Reading Paper: (30%) Write a 2-page summary of a research paper on the application of ML in Chemistry. Describe the problem, approach, why that approach is appropriate, results, and limitations i.e. scopes of improvement.

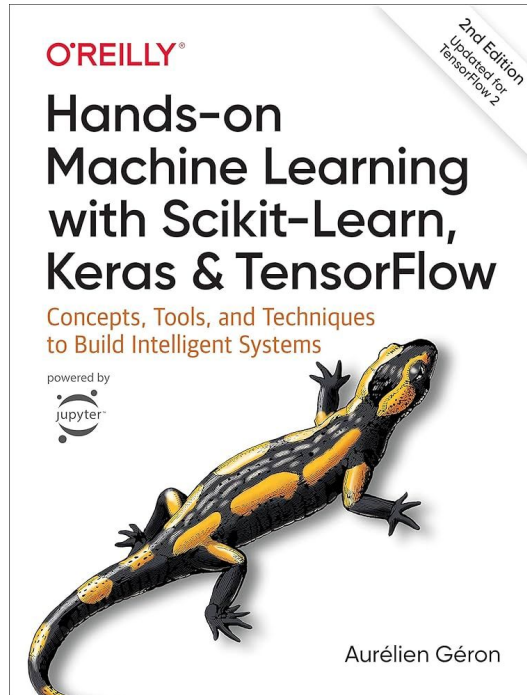
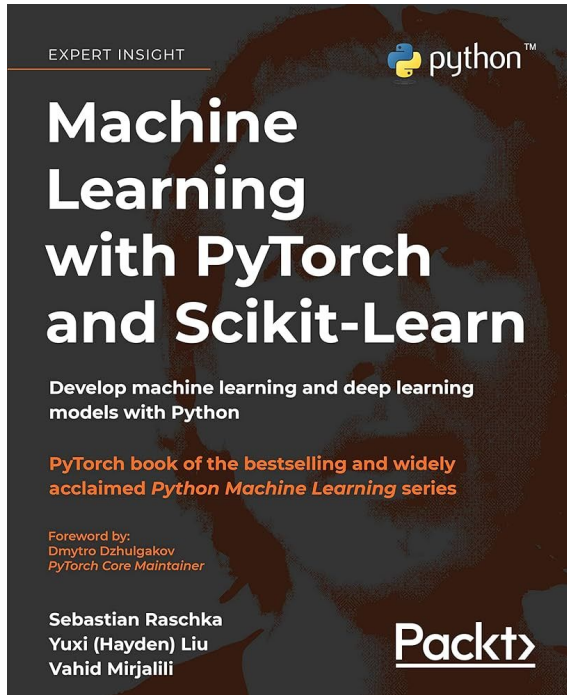
Each lecture includes references application of the ML technique on a chemistry related problem. You can pick any of these papers. You are open to pick any other paper based on your interest, but please check with Dhiman first.

Project: (30%) Application of one (or more) of the ML techniques you learnt in this class on a research problem of your interest. To be submitted in a jupyter notebook format including all codes, plots, workflow and justification of the approach.

Example: An End-to-end Machine Learning Project: Chapter 2 of Geron book.

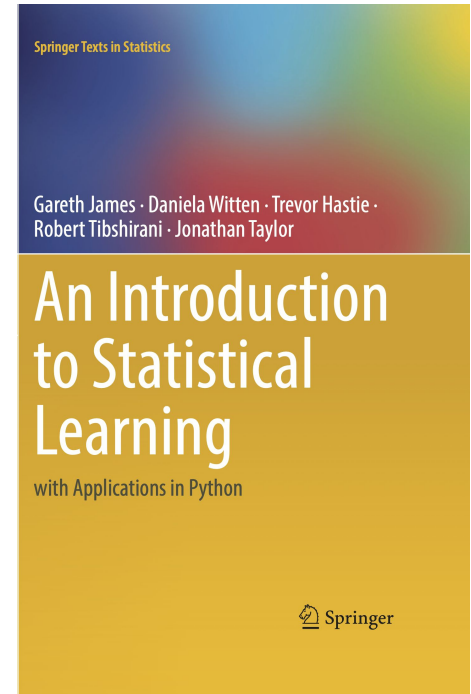
Course Logistics: References

Primary textbooks: available in UO Library



Additional Reference:
Free PDF:

<https://www.statlearning.com>



What is Machine Learning?

Machine Learning is the science (and art) of programming computers so they can learn from data.

[Machine Learning is the] field of study that gives computers the ability to learn without being explicitly programmed.

—Arthur Samuel, 1959

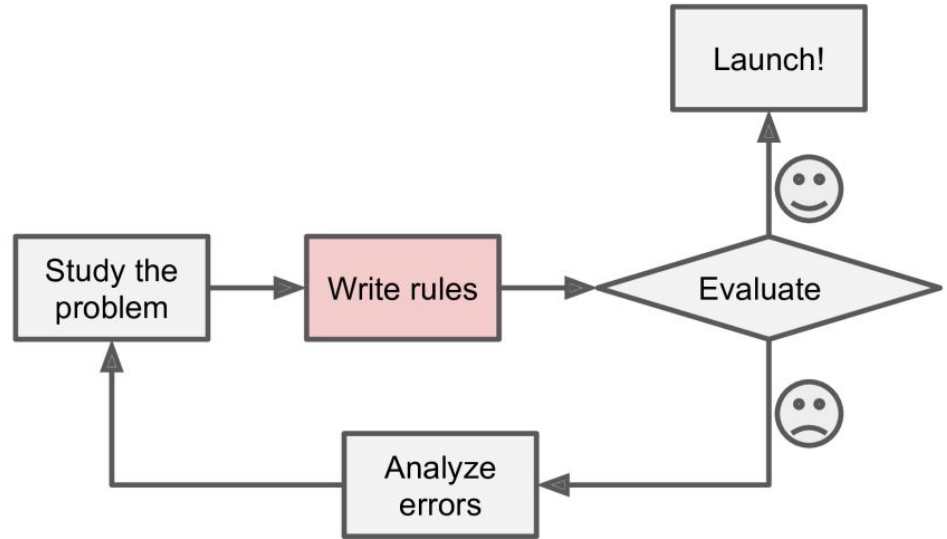
A computer program is said to learn from experience E with respect to some task T and some performance measure P , if its performance on T , as measured by P , improves with experience E .

—Tom Mitchell, 1997

Traditional Learning

Goal: Design a spam filter for emails

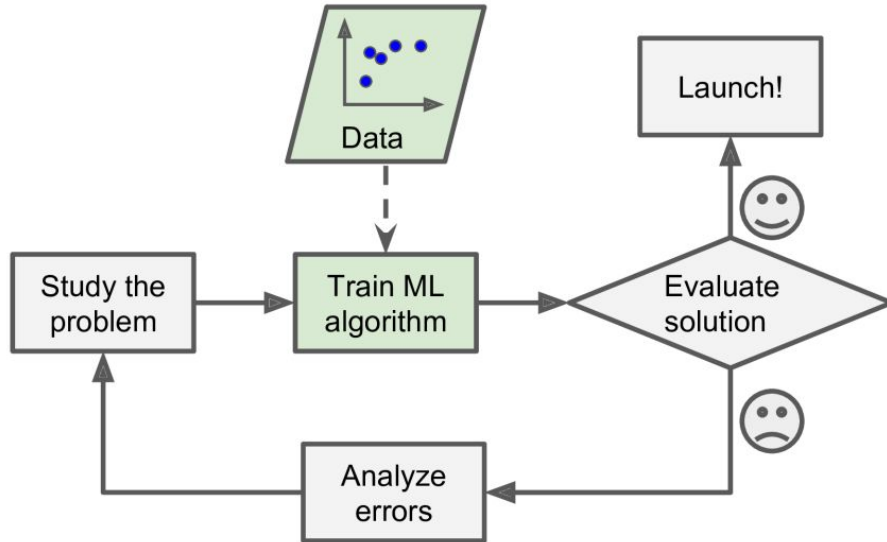
1. Find words or patterns typically appearing on spam emails
2. Write a detection algorithm for each of the patterns that you noticed
3. Test your program, and repeat steps 1 and 2 until it is good enough



- Long list of complex rules: hard to maintain
- Easily overcome by slight changes in email structure

Machine Learning

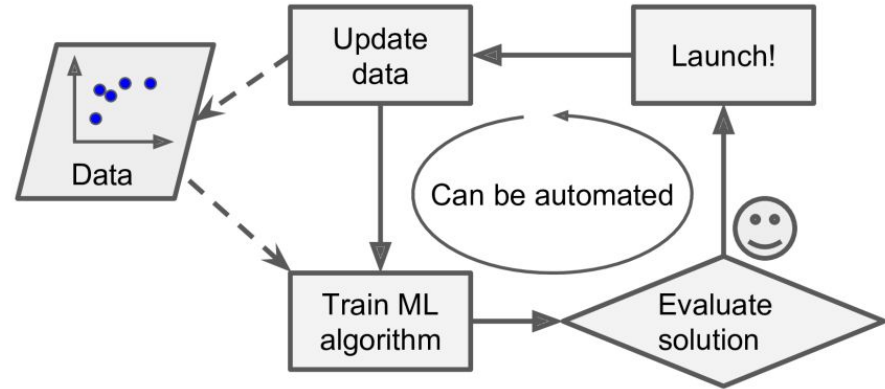
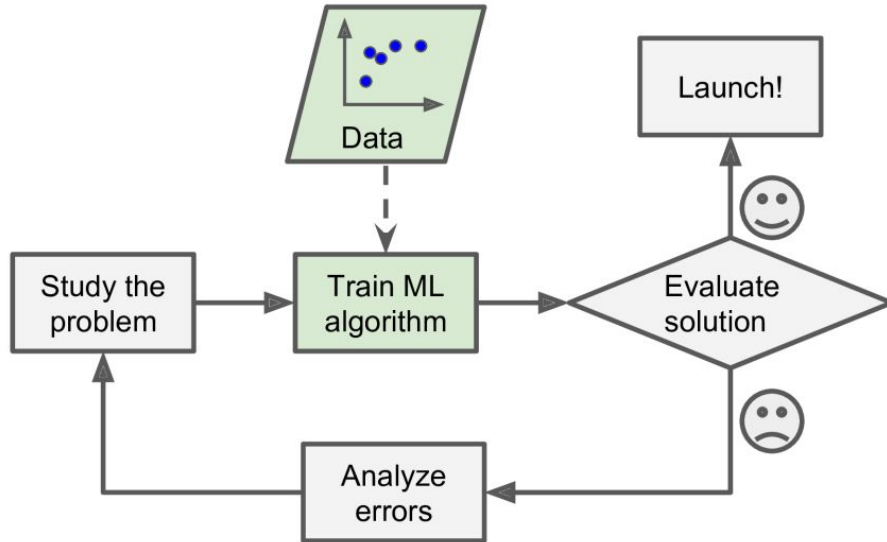
Automatically learns which words and phrases are good predictors of spam by detecting unusually frequent patterns of words in the spam examples



- Program is much shorter, easier to maintain, and most likely more accurate

Machine Learning

Automatically learns which words and phrases are good predictors of spam by detecting unusually frequent patterns of words in the spam examples



- Program is much shorter, easier to maintain, and most likely more accurate
- Can automatically update in response to new data without user intervention

Why Use Machine Learning in Chemistry?

The chemistry disciplines embracing the AI revolution

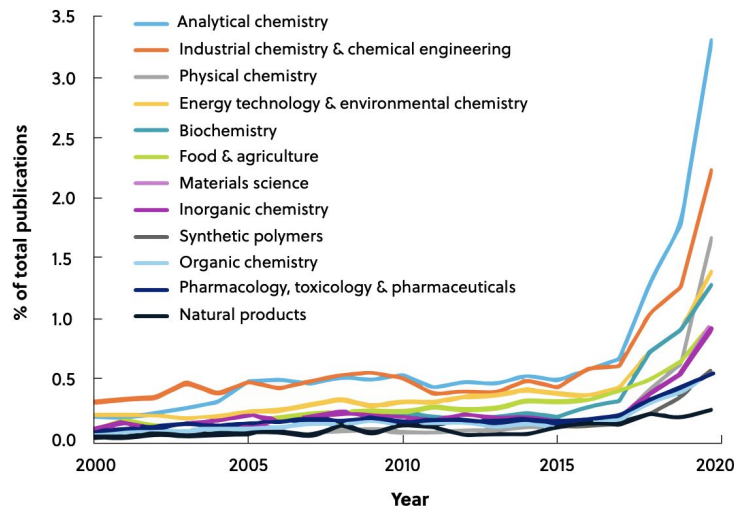


Figure 3. Journal publication trends of AI in specific research areas from 2000 to 2020

With large chemistry datasets being available from experimental research it is important to find ways to interpret and learn from them. Manual learning is too cumbersome from such large datasets.

Why Use Machine Learning in Chemistry?

The chemistry disciplines embracing the AI revolution

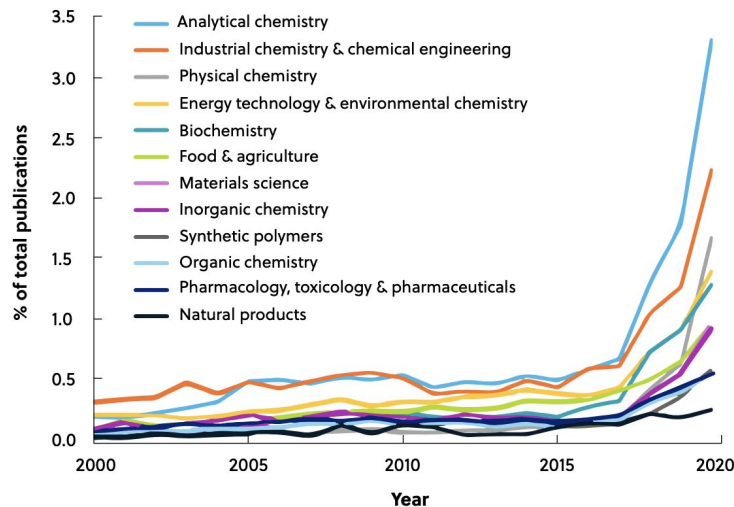
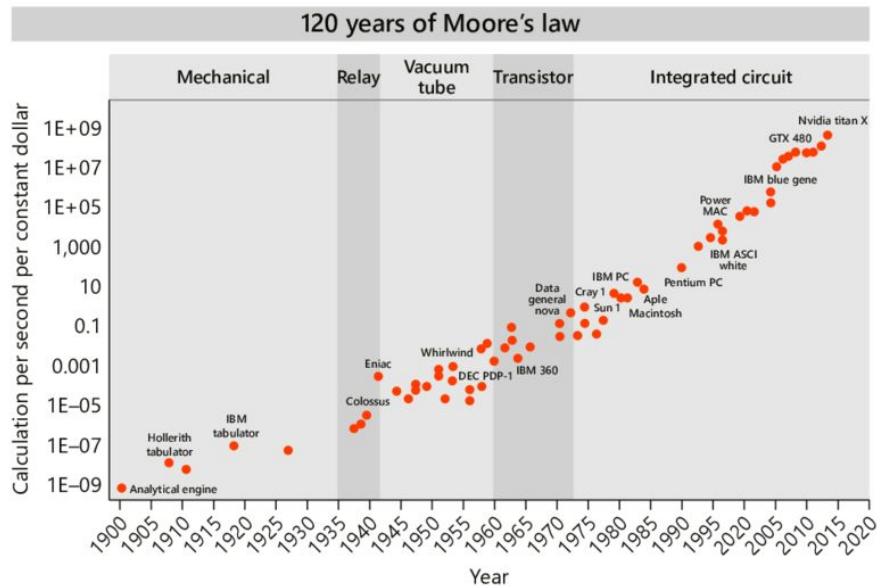


Figure 3. Journal publication trends of AI in specific research areas from 2000 to 2020

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First principles methods (e.g. Quantum Mechanics) are either too expensive or not available at all.

Why Use Machine Learning in Chemistry?



Pea and Cole, *Human Development*
(2019) 62 (1-2): 14–39.

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First principles methods (e.g. Quantum Mechanics) are either too expensive or not available at all.

Increased capacity of computations allow treating big data to learn chemical principle

Availability of Large Datasets (Big Data)

The Nobel Prize in Chemistry 2024



III. Niklas Elmehed © Nobel Prize Outreach

David Baker

Prize share: 1/2



III. Niklas Elmehed © Nobel Prize Outreach

Demis Hassabis

Prize share: 1/4



III. Niklas Elmehed © Nobel Prize Outreach

John Jumper

Prize share: 1/4



Computational Protein Design using
Machine Learning Algorithms

Availability of Large Datasets (Big Data)

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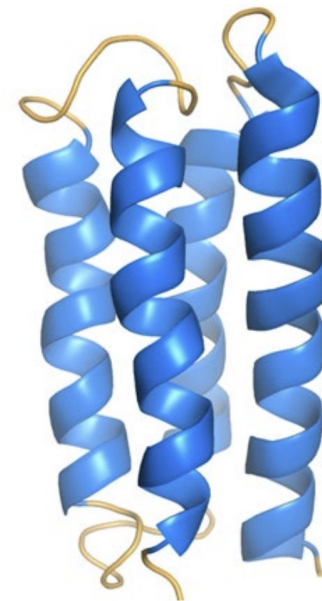
Prize share: 1/4



III. Niklas Elmehed © Nobel Prize Outreach

John Jumper

Prize share: 1/4



Computational Protein Design using
Machine Learning Algorithms

Protein Data
Bank (PDB)

Availability of Large Datasets (Big Data)

The Nobel Prize in Chemistry 2024



Ill. Niklas Elmehed © Nobel Prize Outreach

David Baker

Prize share: 1/2



Ill. Niklas Elmehed © Nobel Prize Outreach

Demis Hassabis

Prize share: 1/4



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Computational Protein Design using Machine Learning Algorithms

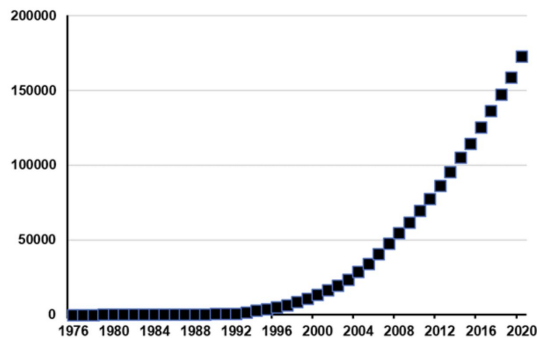
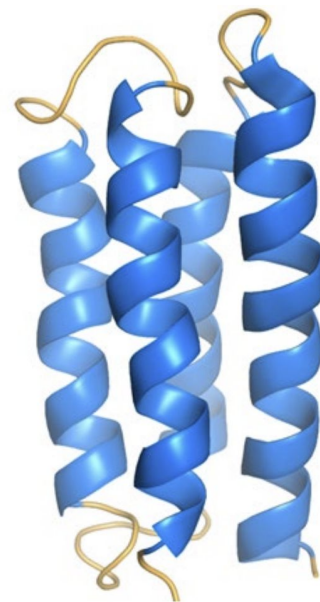
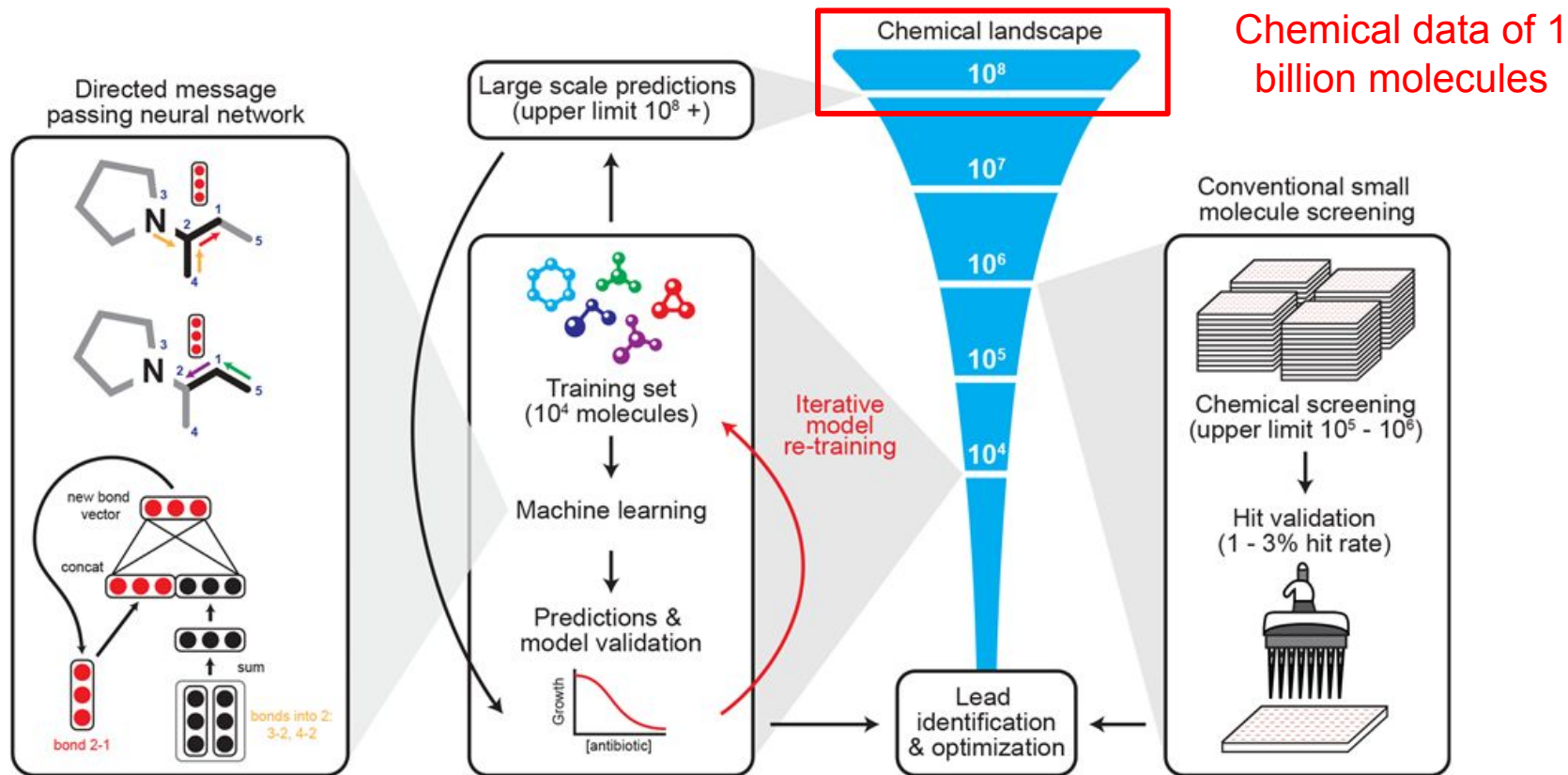


Figure 2. Time evolution of the number of experimentally determined protein structures deposited in the Protein Data Bank.



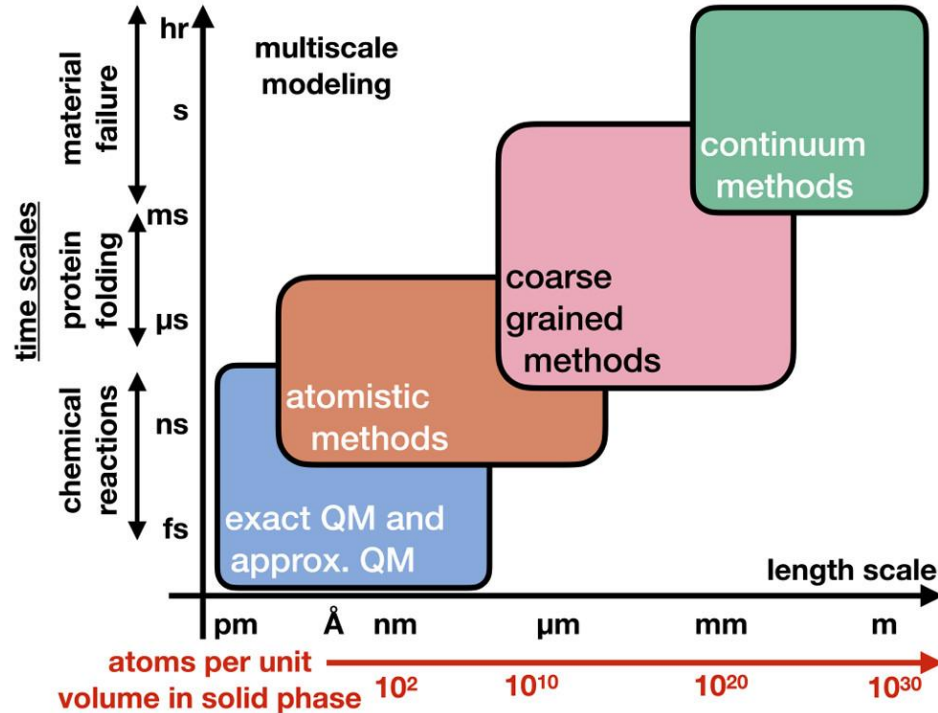
Protein Data Bank (PDB)

Availability of Large Datasets (Big Data)



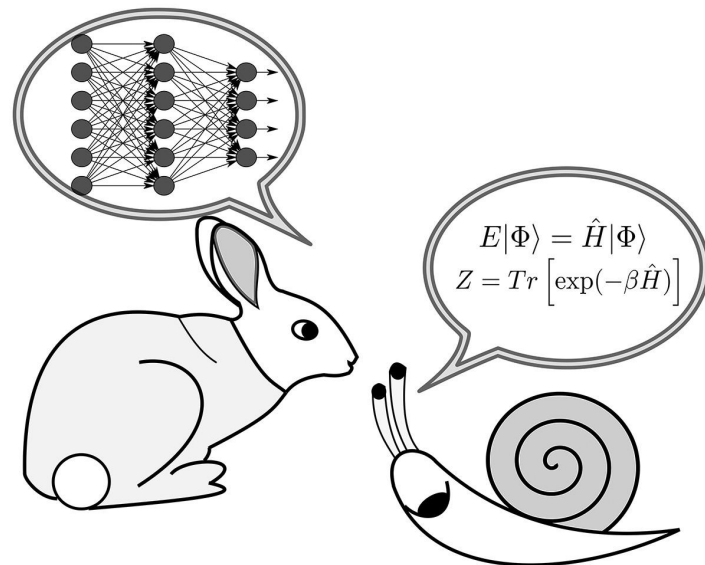
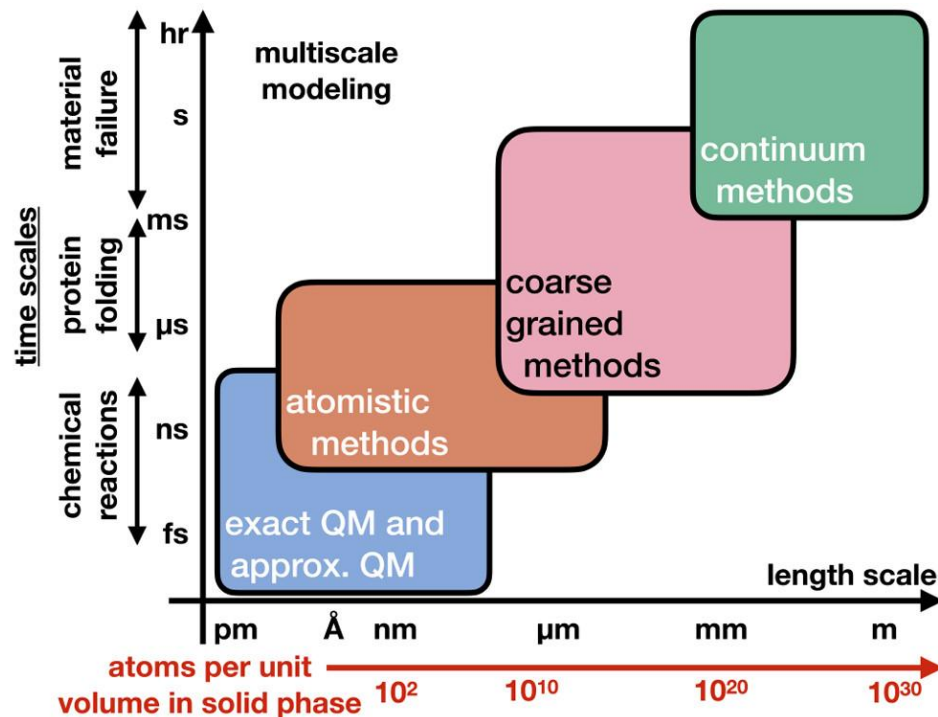
Computational Efficiency

Machine Learning simulations are faster than QM and Molecular Dynamics



Computational Efficiency

Machine Learning simulations are faster than QM and Molecular Dynamics

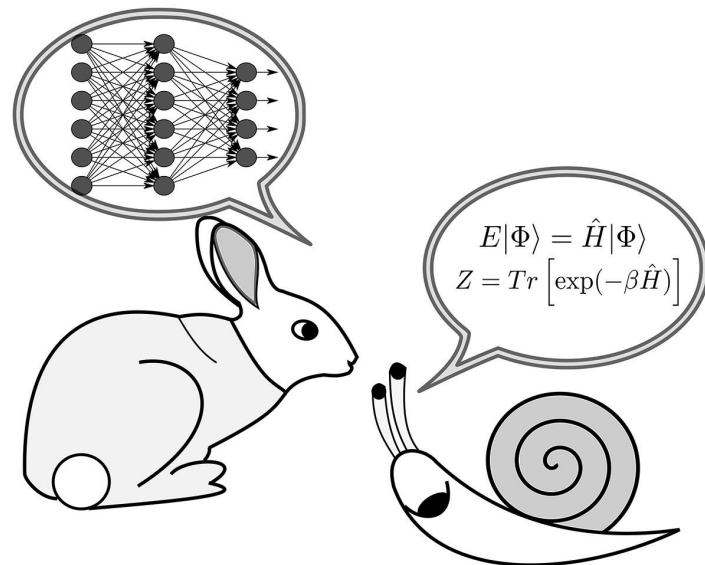
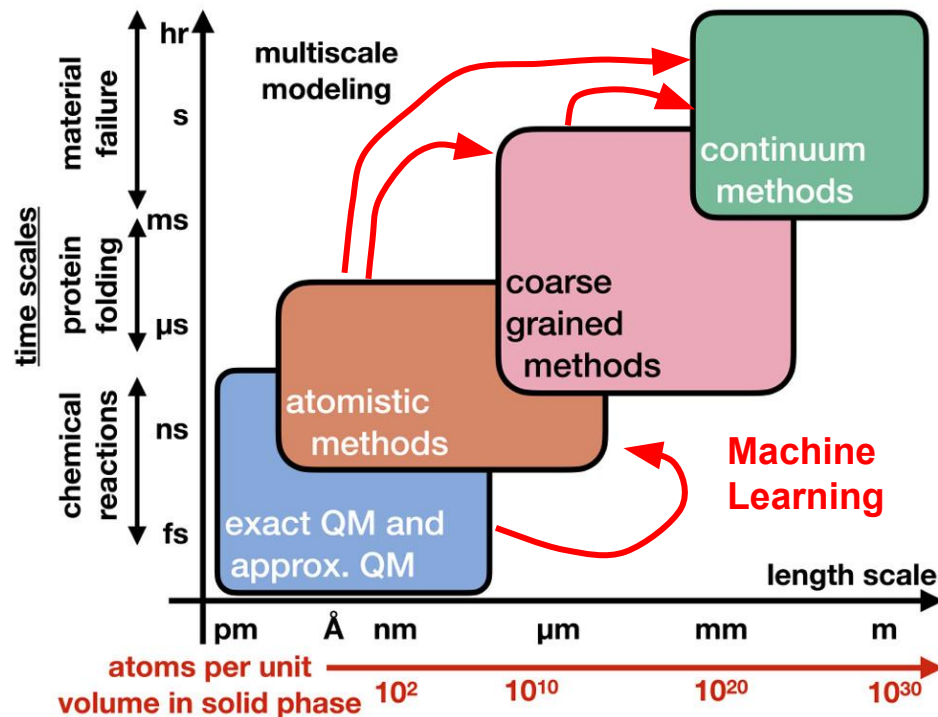


Combining Machine Learning and Computational Chemistry for Predictive Insights Into Chemical Systems

John A. Keith, et al. Chemical Reviews 2021 121 (16), 9816-9872 DOI: 10.1021/acs.chemrev.1c00107

Computational Efficiency

Machine Learning simulations are faster than QM and Molecular Dynamics



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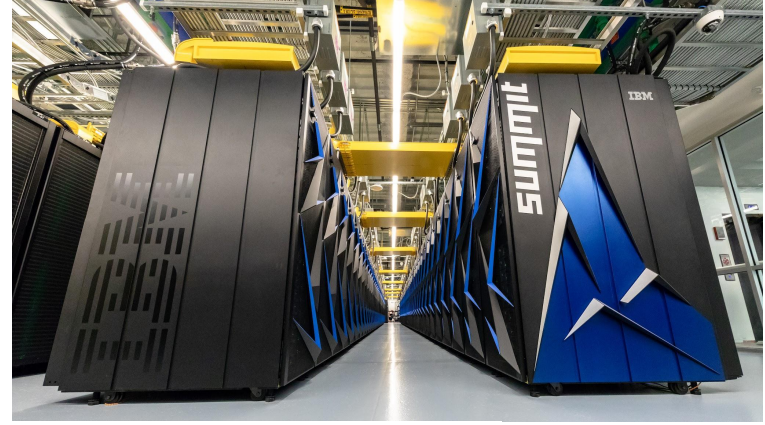
Revolution in Parallel Computing



**HPC Cluster
(~\$1M)**

<- SDSC
(San Diego)

Summit ->
(ORNL)



**Graphical Processing Unit (GPU)
Workstations
(~\$5-10k)**

Affordable for small research groups

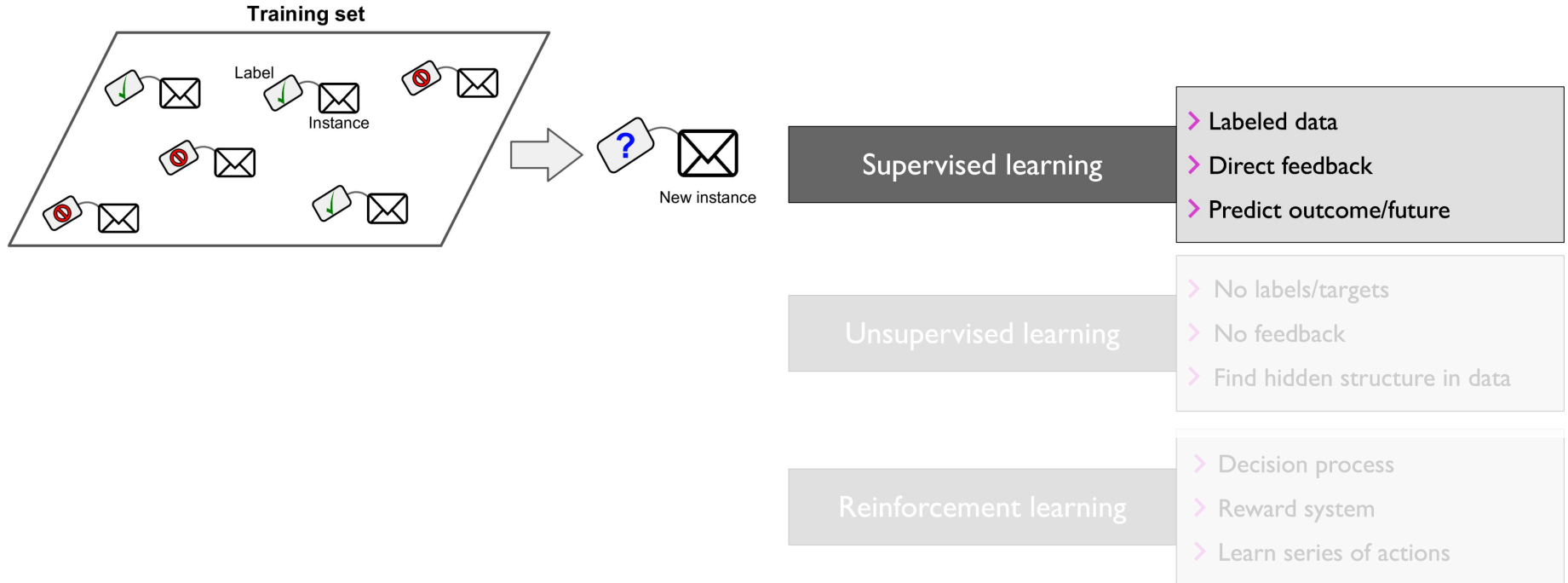


Types of Machine Learning

Supervised learning	➤ Labeled data ➤ Direct feedback ➤ Predict outcome/future
Unsupervised learning	➤ No labels/targets ➤ No feedback ➤ Find hidden structure in data
Reinforcement learning	➤ Decision process ➤ Reward system ➤ Learn series of actions

Types of Machine Learning

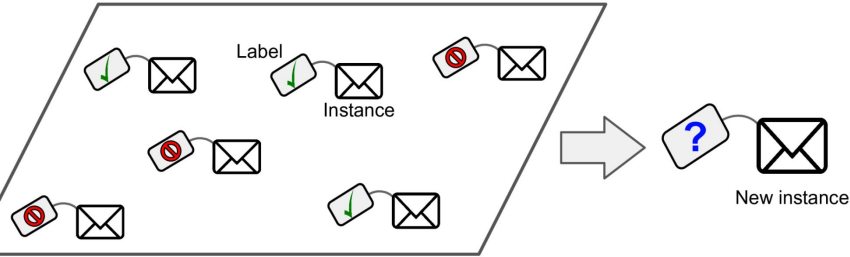
Supervised Learning: Predicting outcome of a problem



Types of Machine Learning

Supervised Learning: Predicting outcome of a problem

Training set



Supervised learning

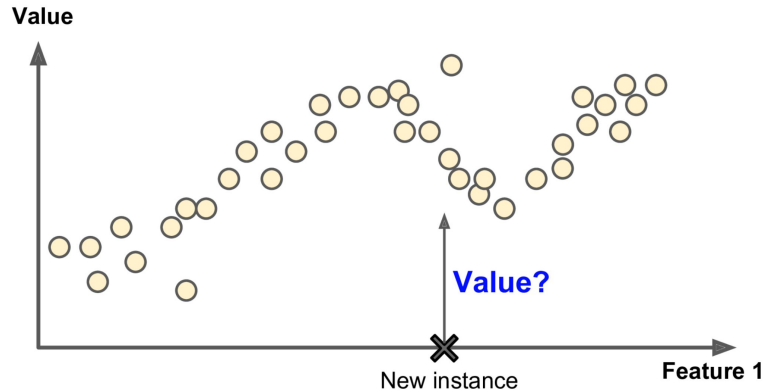
- > Labeled data
- > Direct feedback
- > Predict outcome/future

Unsupervised learning

- > No labels/targets
- > No feedback
- > Find hidden structure in data

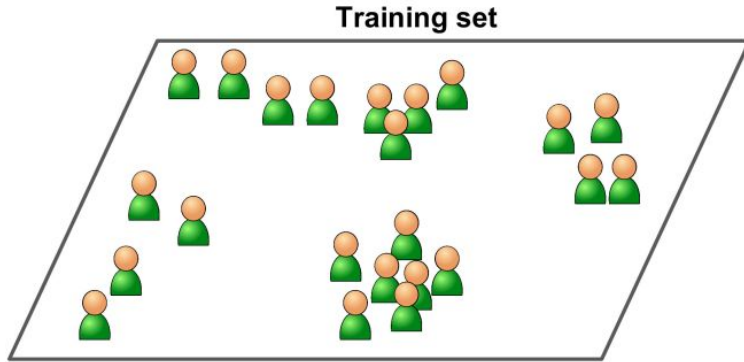
Reinforcement learning

- > Decision process
- > Reward system
- > Learn series of actions



Types of Machine Learning

Unsupervised Learning: Finding patterns in unlabelled training data.



Supervised learning

- > Labeled data
- > Direct feedback
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Unsupervised learning

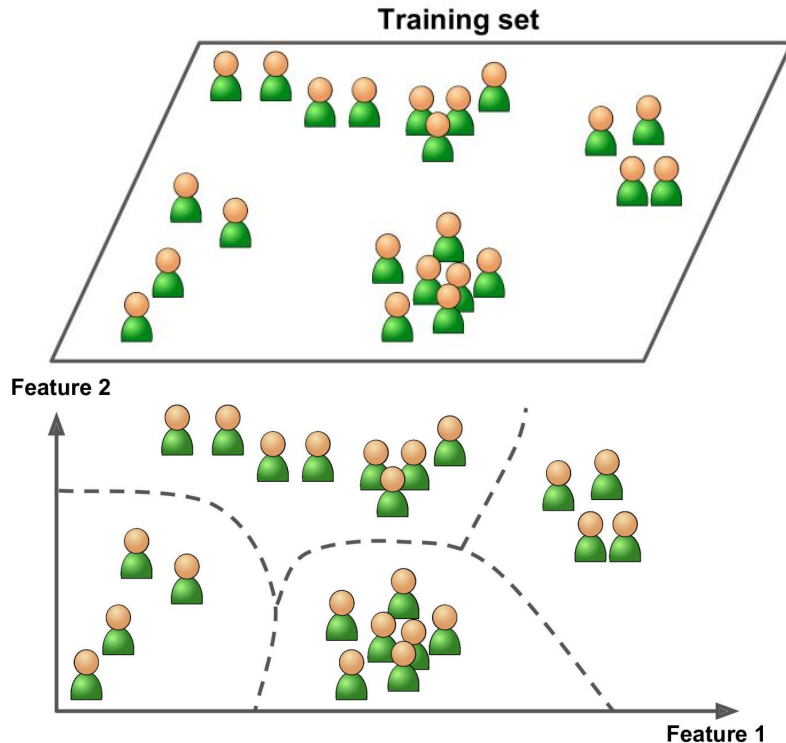
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Types of Machine Learning

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Unsupervised learning

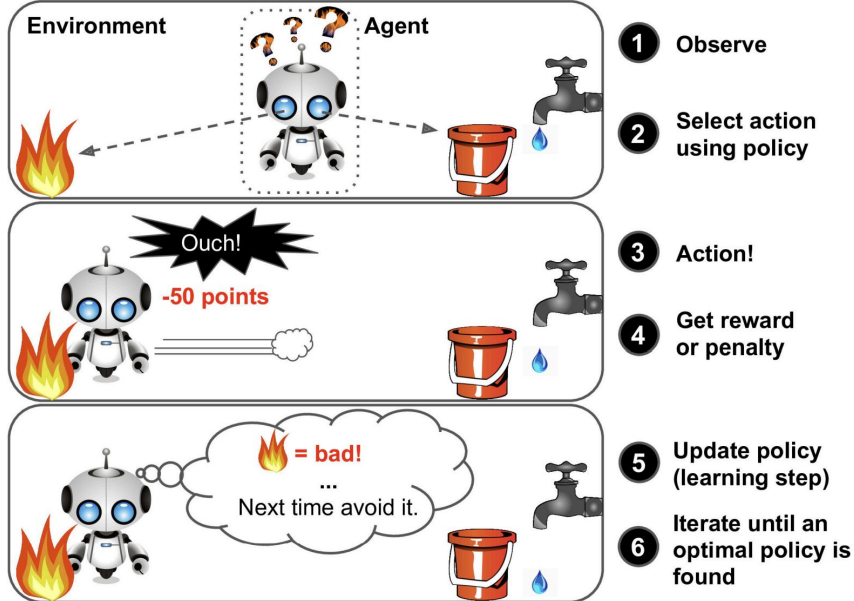
- > No labels/targets
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Reinforcement learning

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- > Reward system
- > Learn series of actions

Types of Machine Learning

Reinforcement Learning: Learn from actions (prize/punishment)



Supervised learning

- > Labeled data
- > Direct feedback
- > Predict outcome/future

Unsupervised learning

- > No labels/targets
- > No feedback
- > Find hidden structure in data

Reinforcement learning

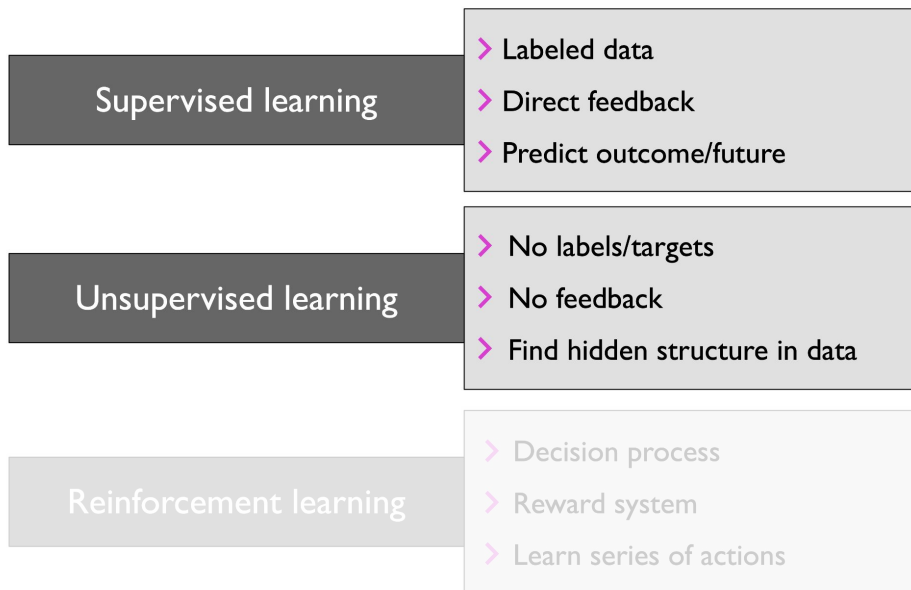
- > Decision process
- > Reward system
- > Learn series of actions

Types of Machine Learning

This course will focus on Supervised and Unsupervised Learning.

More suitable for chemical problems e.g. predicting molecular property, reactivity etc.

You can learn more about reinforcement learning from the two textbooks.



Introduction to Python Programming